

Faithful Extraction of Citations from Academic PDFs

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Abstract

We describe a pipeline that extracts positioned text from academic PDFs, orders it into columns, and resolves in-text citation markers against the reference list.

1 Introduction

Academic PDFs encode text as positioned glyph runs rather than as a reading-order stream. Faithful extraction must recover the columns before any downstream parsing can succeed [1]. Prior systems relied on heuristics tuned to single-column layouts. We instead treat every span as evidence and keep the raw bytes of each reference entry.

Our contribution is a pipeline that snapshots the input, extracts positioned spans with a pure Rust backend, orders them with a recursive XY-cut, and resolves numeric citation markers such as [2, 3] against the segmented reference list. Every field that we report carries provenance, and nothing is repaired or guessed by a language model.

2 Method

The reference list is located by its heading and segmented into entries using the printed labels and the hanging indent of each continuation line, following Turing [3]. Each entry keeps its raw text; parsed fields are best-effort and never invented or repaired.

References

- [1] A. Lovelace and C. Babbage, "Analytical engines," in Proceedings of the 1st Conference on Computing, 1843, pp. 1–10.
- [2] G. Hopper, "Compilers," Journal of Programming, vol. 3, no. 2, pp. 45–67, 1952, doi:10.1000/abc456.
- [3] A. Turing, "On computable numbers," arXiv:1936.00001, 1936.